ANALYSIS OF A DIELECTRIC CHANNEL WAVEGUIDE DIFFUSED IN AN ANISOTROPIC SUBSTRATE WITH A GAUSSIAN-GAUSSIAN INDEX OF REFRACTION PROFILE USING THE FINITE-DIFFERENCE METHOD.

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ABSTRACT

A dielectric channel waveguide diffused in an isotropic (or anisotropic) substrate with an index of refraction profile varying arbitrarily in the waveguide cross section is analyzed here using a rigorous formulation of the finite-difference method. In this formulation the finite-difference method is used in the numerical solution of the wave equation written in terms of the transverse components of the magnetic field. As a result, a conventional eigenvalue problem is obtained. The elimination of the spurious modes (observed in previous formulations) is obtained by the implicit inclusion of the divergence of the magnetic field equal to zero. The formulation is general and may be applied in the analysis of various structures of practical interest such as in millimeter and optical integrated circuits.

I. INTRODUCTION

The finite-difference method, developed in terms of transverse components of the magnetic field, has been initially used in the analysis of a channel dielectric waveguide diffused in an isotropic substrate [1], [2].

In this paper the formulation developed by Bierwirth et al. [1] and Schulz et al. [2] is extended to include biaxial anisotropic dielectrics with a refractive index profile varying arbitrarily in the waveguide cross section. The formulation was first tested for waveguides with a Gaussian-Gaussian index of refraction profile in the x and y variables [3] and then for channel dielectric waveguides diffused in anisotropic substrate [4]. A good agreement between results from [3] and [4] and our results is observed.

II. THEORY

In the formulation described here the anisotropic dielectrics are oriented such that the optical axes coincide with the coordinate system axes, as shown in Fig. 1. As a result the permittivity tensor of the biaxial anisotropic dielectric may be written as

$$[\varepsilon] = \begin{bmatrix} \varepsilon_x(x,y) & 0 & 0 \\ 0 & \varepsilon_y(x,y) & 0 \\ 0 & 0 & \varepsilon_{(z)}(x,y) \end{bmatrix} \tag{1}$$

The magnetic permeability of the dielectrics is assumed equal to the free space permeability, μ_o. The fields have a harmonic time dependence, $\exp(j\omega t)$, and propagate along the z direction (see Fig. 1), with a dependence $\exp(-\gamma_z z)$, where ω is the angular frequency and γ_z is the propagation constant.

The vector wave equation for a cylindrical waveguide with a non-homogeneous cross section and anisotropic dielectric material may be obtained from Maxwell's equations as

$$-[\varepsilon]^{-1}\nabla^2\,\overline{H} + [\nabla([\varepsilon]^{-1})] \times (\nabla \times \overline{H}) = \omega^2 \mu\,\overline{H}. \tag{2}$$

With the objective of simplifying the problem solution, (2) may be written in terms of the transverse components of the magnetic field, H_x and H_y. As a result, the following coupled equations are obtained [5]

$$\frac{\varepsilon_\alpha(x,y)}{\varepsilon_z(x,y)}\frac{\partial^2}{\partial \alpha^2}H_\tau + \frac{\partial^2}{\partial \tau^2}H_\tau + \frac{\varepsilon_z(x,y)-\varepsilon_\alpha(x,y)}{\varepsilon_z(x,y)}\frac{\partial^2}{\partial \tau\,\partial \alpha}H_\alpha$$
$$+ \frac{\varepsilon_\alpha(x,y)}{\varepsilon_z^2(x,y)}\frac{\partial}{\partial \alpha}\varepsilon_z(x,y)\left(\frac{\partial}{\partial \tau}H_\alpha - \frac{\partial}{\partial \alpha}H_\tau\right) + \left(k_o^2\frac{\varepsilon_\alpha(x,y)}{\varepsilon_o} + \gamma_z^2\right)H_\tau = 0, \tag{3}$$

where $\alpha = x$ when $\tau = y$, $\alpha = y$ when $\tau = x$, ε_o is the free-space dielectric permittivity and $k_o = \omega(\mu_o\varepsilon_o)^{1/2}$ is the free-space wavenunber.

With the application of the coupled equations (3) in the four regions, 1, 2, 3 and 4 of the five points graded mesh of Fig. 2, eight equations are obtained. Besides these eight equations, a group of equations obtained by imposing the continuity condition of the longitudinal components of the electric and magnetic fields, E_z and H_z, at the interfaces 1-2, 2-3, 3-4 and 4-1 of Fig. 2, are also included. From the various possible continuity equations groups we have chosen for H_x : $E_{z1} = E_{z2}$, $E_{z3} = E_{z4}$, $H_{z1} = H_{z2}$, $H_{z2} = H_{z3}$, $H_{z3} = H_{z4}$, and for H_y: $E_{z1} = E_{z4}$, $E_{z2} = E_{z3}$, $H_{z1} = H_{z4}$, $H_{z2} = H_{z3}$, $H_{z1} = H_{z2}$. We thus obtain the desired solution of the wave equation for the point P of Fig. 2. The result is then applied to each mesh point of Fig. 1 [5]. By taking into consideration the boundary conditions at the surfaces that limit the region of interest, the problem is reduced into a conventional eigenvalue problem, with the elimination of the spurious modes due to the implicit inclusion of $\nabla \cdot \bar{H} = 0$. The resulting expression is [5]

$$[(A) - \lambda (U)](X) = 0, \qquad (4)$$

where (A) is a band matrix containing constant coefficients, (U) is the unit matrix, (X) is a column vector, containing the transverse components of the magnetic field, H_x and H_y, and λ is the eigenvalue that gives the propagation constant, γ_z, of the guided modes ($\lambda = -\gamma_z^2$). The solution of (4) was obtained in an IBM 3090 computer using the EISPACK program.

III. RESULTS AND CONCLUSIONS

The dispersion characteristics of dielectric channel waveguides diffused in an isotropic (or anisotropic) substrate with a Gaussian-Gaussian refractive index profile were examined using a rigorous finite-difference method.

The dispersion characteristics of the E^y_{11} and E^y_{21} modes of an anisotropic dielectric channel waveguide diffused in Li Nb O_3 are shown in Fig. 3. The results obtained with this analysis are given by a solid line and they agree well with the results from Koshiba et al. [4] (shown by dots), obtained using the scalar finite-element method.

Fig. 4 shows the dispersion characteristics of the normalized phase constant, B, as a function of the normalized frequency, V, with

$$B = [(\beta_z / k_o) - n_2^2] / (n_{3max}^2 - n_2^2) \text{ and } V = k_o b \sqrt{n_{3max}^2 - n_2^2}$$

where β_z is the phase constant, n_2 is the substrate index of refraction, n_{3max} is the maximum value of the index of refraction in the channel and b is one of the waveguide transverse dimensions (see Fig. 1), for the E^y_{11} and E^y_{12} of an isotropic dielectric channel waveguide with a bidimensional refractive index varying in a Gaussian-Gaussian form. The results obtained with this formulation (solid line) show a good agreement with those represented by dots, obtained by Lagu and Ramaswamy [3] using a variational finite-difference method.

Figs. 5, 6 and 7 show the dispersion characteristics of the E^x_{pq} and E^y_{pq} modes for a dielectric channel waveguide diffused in a dielectric substrate. In Fig. 5 the substrate is isotropic, while in Figs. 6 and 7 the substrate is an uniaxial anisotropic dielectric. In Fig. 6 the optical axis is oriented along the y direction, while in Fig. 7 it is along the x direction.

Examining Figs. 5, 6 and 7 we observe that, the introduction of the anisotropy causes a displacement of the modes with an electric field component along the optical axis. In Fig. 6 this happens to the E^y_{pq} modes while in Fig. 7 this happens to the E^x_{pq} modes. Therefore, in Fig. 6, a single E^x_{11} mode operation is possible from its cutoff frequency to the cutoff frequency of the next higher order mode (E^x_{21}) . Similar observation is made in Fig. 7. For the higher order modes and at the higher frequencies the convergence of the numerical results require finer meshes.

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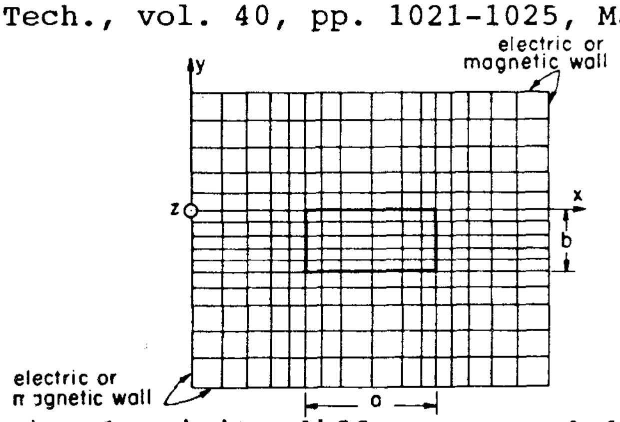

Fig. 1. Finite-difference graded mesh representation.

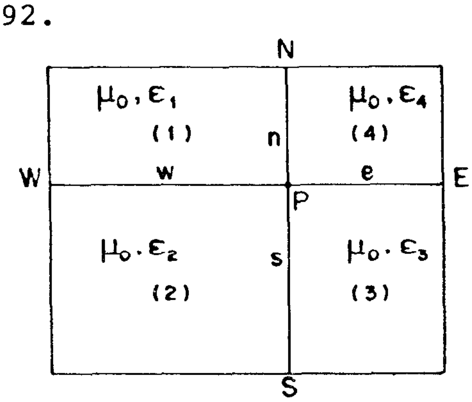

Fig. 2. Five-points graded mesh representation.

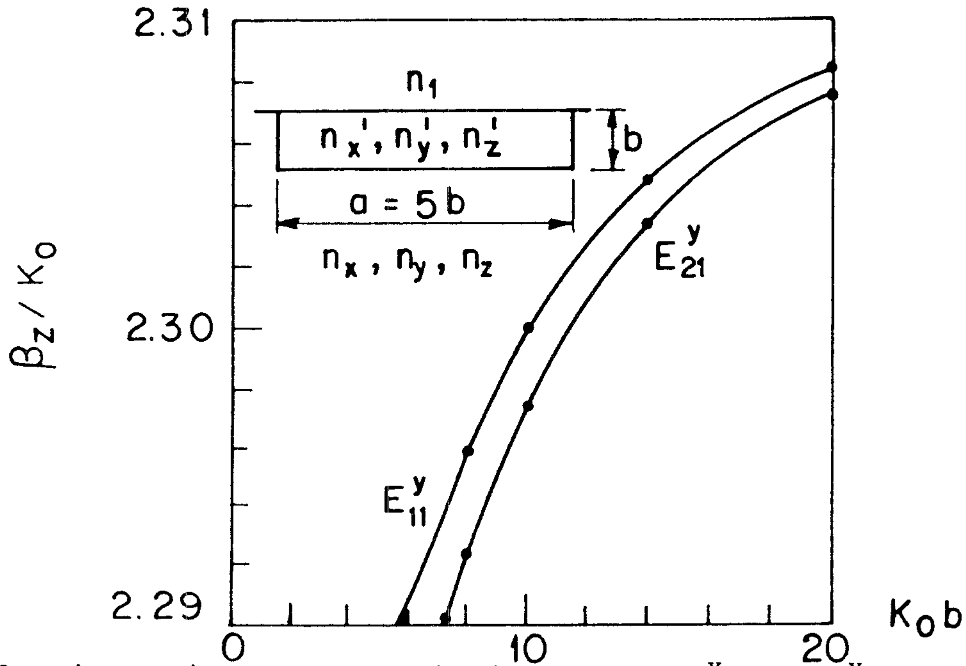

Fig. 3. Dispersion characteristics of the E^y_{11} and E^y_{21} modes of an anisotropic dielectric channel waveguide diffused in $LiNbO_3$. Results from this analysis are shown by solid lines while the dots are results obtained by Koshiba et al. [4]. $n_x = 2.2$, $n_y = n_z = 2.29$, $n_x' = 2.222$, $n_y' = n_z' = 2.3129$, $n' = 1.0$.

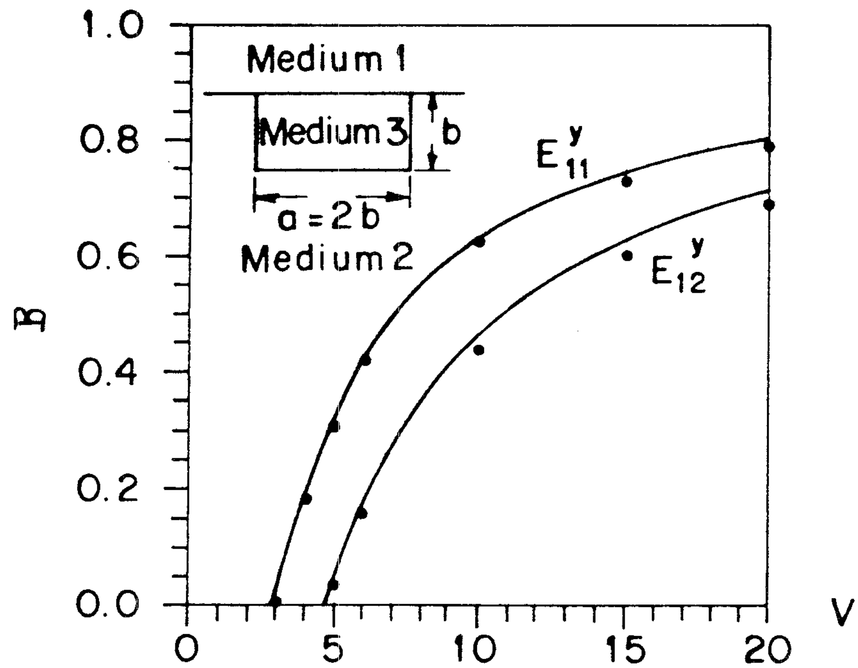

Fig. 4. Dispersion characteristics of the E^y_{11} and E^y_{12} modes of a diffused dielectric channel waveguide. The results from this analysis are shown by solid lines while the dots are results obtained by Koshiba et al. [4]. $n_y = 1.0$, $n_2 = 2.1$, $n_3 = n_2 \cdot [1 + 0.05 f(x,y)]$, where $f(x,y) = \exp[-4(x - x_o)^2 / a^2] \cdot \exp[-(y/b)^2]$ and x_o is the abscissa of the guide central point.

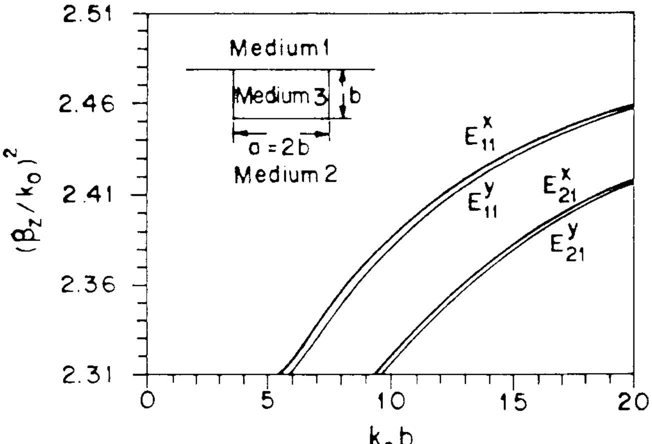

Fig. 5. Dispersion characteristics of the E^x_{pq} and E^y_{pq} modes of a diffused dielectric channel waveguide. $n_1 = 0$, $n_2 = \sqrt{2.31}$, $n_3 = n_2[1 + 0.05f(x,y)]$ with $f(x,y)$ given in Fig. 4.

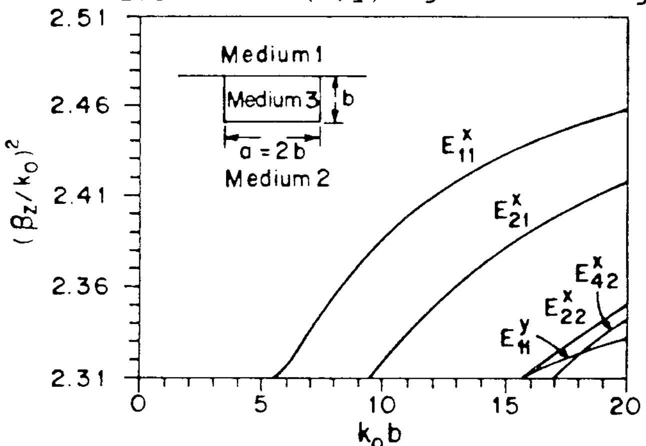

Fig. 6. Dispersion characteristics of the E^x_{pq} and E^y_{pq} modes of a diffused dielectric channel waveguide. $n_1 = 1.0$, $n_{2x} = n_{2z} = \sqrt{2.31}$, $n_{2y} = \sqrt{2.19}$, $n_{3y} = n_{2y} + 0.05\, n_{2x}\, f(x,y)$, and $n_{3x} = n_{3z} = n_{2x}[1 + 0.05f(x,y)]$, with $f(x,y)$ given in Fig. 4.

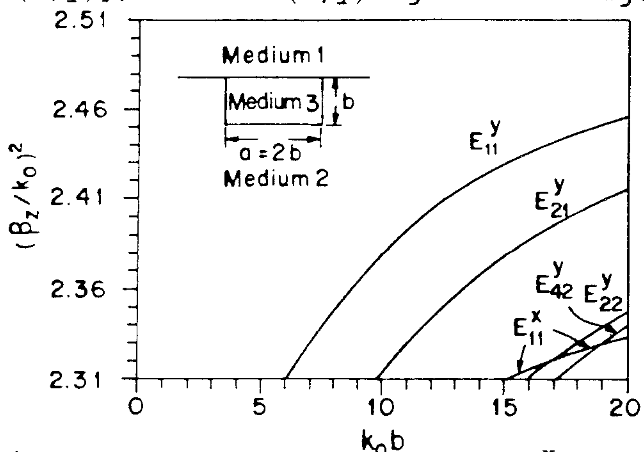

Fig. 7. Dispersion characteristics of the E^x_{pq} and E^y_{pq} modes of a diffused dielectric channel waveguide. $n_1 = 1.0$, $n_{2x} = \sqrt{2.19}$, $n_{2y} = n_{2z} = \sqrt{2.31}$, $n_{3x} = n_{2x} + 0.05\, n_{2y}\, f(x,y)$, and $n_{3y} = n_{3z} = n_{2y}[1 + 0.05f(x,y)]$, with $f(x,y)$ given in Fig. 4.